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**The Logic of Transgenerational
 Inheritance: Timescales of
 Adaptation**

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Keywords

transgenerational epigenetic inheritance, TEI, intergenerational
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Abstract

Myriad mechanisms have evolved to adapt to changing environments. Environmental stimuli alter organisms' physiology to create memories of previous environments. Whether these environmental memories can cross the generational barrier has interested scientists for centuries. The logic of passing on information from generation to generation is not well understood. When is it useful to remember ancestral conditions, and when might it be deleterious to continue to respond to a context that may no longer exist? The key might be found in understanding the environmental conditions that trigger long-lasting adaptive responses. We discuss the logic that biological systems may use to remember environmental conditions. Responses spanning different generational timescales employ different molecular machineries and may result from differences in the duration or intensity of the exposure. Understanding the molecular components of multigenerational inheritance and the logic underlying beneficial and maladaptive adaptations is fundamental to understanding how organisms acquire and transmit environmental memories across generations.

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INTRODUCTION

Transgenerational epigenetic inheritance (TEI) refers to an unconventional, nongenetic form of inheritance whereby an organism's experiences are encoded in the form of epigenetic changes and persist over multiple generations, even in the absence of the original stimulus. Epigenetic inheritance can be mediated by enzymatic modifications to DNA, chromatin modifications, and noncoding RNAs (Fitz-James & Cavalli 2022, Trerotola et al. 2015). These modifications may be induced by environmental stimuli (Kaletsky et al. 2020; Klosin et al. 2017; Mondotte et al. 2020; Moore et al. 2019, 2021a; Rasmann et al. 2012; Toker et al. 2022; Webster et al. 2018) or by endogenous regulatory programs (Cubas et al. 1999), and they are usually inherited for a set number of generations. TEI is widespread in some plant species that can epigenetically adapt to several environmental stressors, passing on the resulting adaptive changes to successive generations, usually in a DNA methylation-dependent manner (Rasmann et al. 2012, Rendina González et al. 2018). In invertebrates like *Drosophila* and *Caenorhabditis elegans*, parental diet and stress can have long-lasting effects on multiple generations of progeny (Hibshman et al. 2016, Moore et al. 2019, Rechavi et al. 2014, Stern et al. 2012, Wan et al. 2022, Wang et al. 2022). Epigenetic inheritance has also been documented in mammals, particularly in the context of diet-related stimuli (Carone et al. 2010), but most of these effects are intergenerational, i.e., lasting only one to two generations. In this review, we focus on the environmental signals that induce TEI and the molecular mechanisms and intertissue signaling downstream of these signals that actuate epigenetic inheritance. We discuss the evolutionary significance of TEI, the nature of epigenetic inheritance across different plant and animal species, and the potential dependence of TEI on the environment, lifestyle, and life history traits.

Whether memories of an organism's ancestor's previous experiences can cross the generational barrier has been studied for centuries. This idea embodies the concepts put forth by Jean-Baptiste Lamarck and others in his theory of inheritance of acquired characteristics (Burkhardt 2013, Zirkle 1935). In a massive experimental effort, August Weismann examined the validity of Lamarck's theory, testing whether altered somatic processes can be inherited through the germ

line. Weismann cut the tails of 901 mice and their progeny for five generations; the progeny of the amputee mice still grew tails, contradicting the idea that acquired characteristics can be inherited. Based on this observation, Weismann proposed the germ-plasm theory—the idea that the germ line serves as a barrier for the inheritance of somatic characteristics (S. 1893). Weismann’s observations and the increasing number of examples of genetic inheritance that corroborated Charles Darwin’s theory starkly reduced support for Lamarck’s theory of acquired characteristics. However, in recent decades, there have been many observations, across plant and animal systems, demonstrating the inheritance of parental experiences across one or more generations (Burton & Greer 2022, Perez & Lehner 2019). In fact, data from historical events, such as famines, and from populations experiencing generational trauma, including Holocaust survivors (Yehuda et al. 2016) and Native American populations (Ehlers et al. 2013), suggest that humans may also exhibit the multigenerational inheritance of parental experiences. It is important to understand the mechanisms and significance of multigenerational inheritance, with the goal of developing effective interventions to combat the inheritance of potentially maladaptive traits.

Decades of research using plants and invertebrate model organisms have documented instances of the multigenerational inheritance of acquired traits (Burton et al. 2021, 2020; Gammon et al. 2017; Guzman et al. 2022; Jobson et al. 2015; Kaletsky et al. 2020; Ma et al. 2019; Moore et al. 2019; Palominos et al. 2017; Perez & Lehner 2019; Perez et al. 2021; Rasmann et al. 2012; Rendina González et al. 2018; Willis et al. 2021). While the underlying molecular mechanisms are beginning to be understood (Fitz-James & Cavalli 2022), many questions remain about the precise molecular logic of acquired traits and how they may be passed on from generation to generation. That is, when is it useful to remember ancestral conditions, and when might it be deleterious to continue to respond to a context that may no longer exist? The key to these questions may be found in understanding the environmental conditions that trigger long-lasting adaptive responses as well as in the differences among species in molecular machinery, physiology, generation times, and life course trajectories. Here we focus on the temporal logic of TEI and ask what determines whether organisms employ intragenerational adaptation, the intergenerational inheritance of adaptive effects, or the transgenerational inheritance of adaptive effects (**Figure 1**) (see the sidebar titled Definition of TEI and Distinction from Intergenerational Inheritance for differences between types of generational inheritance). We outline the possible roles of organismal life histories and different types of environmental stimuli in determining the nature of adaptation (intragenerational, intergenerational, or transgenerational), and in those contexts, we analyze some of the current evidence for transgenerational inheritance in mammals.

TEMPORAL LOGIC OF MULTIGENERATIONAL INHERITANCE: GENERATIONAL SCALES OF ADAPTATION

Intragenerational Responses

Organisms may undergo plastic changes in their physiology in response to dynamic environmental conditions (Xue & Leibler 2018). Some physiological changes are limited to an individual organism’s life, while others can be either passed down to immediate progeny or extended across several generations (Baugh & Day 2020). While the mechanisms governing the rules for the duration of a physiological response are not entirely known, it is intriguing that responses of these varying durations often employ shared molecular machinery (Burton & Greer 2022, Fitz-James & Cavalli 2022, Sarkies 2020), including small RNA (sRNA) pathways and epigenetic regulators. Nonetheless, it is likely that the response duration is not random, but instead it is a function of the type of adaptation. The costs and benefits of multigenerational inheritance are discussed below.

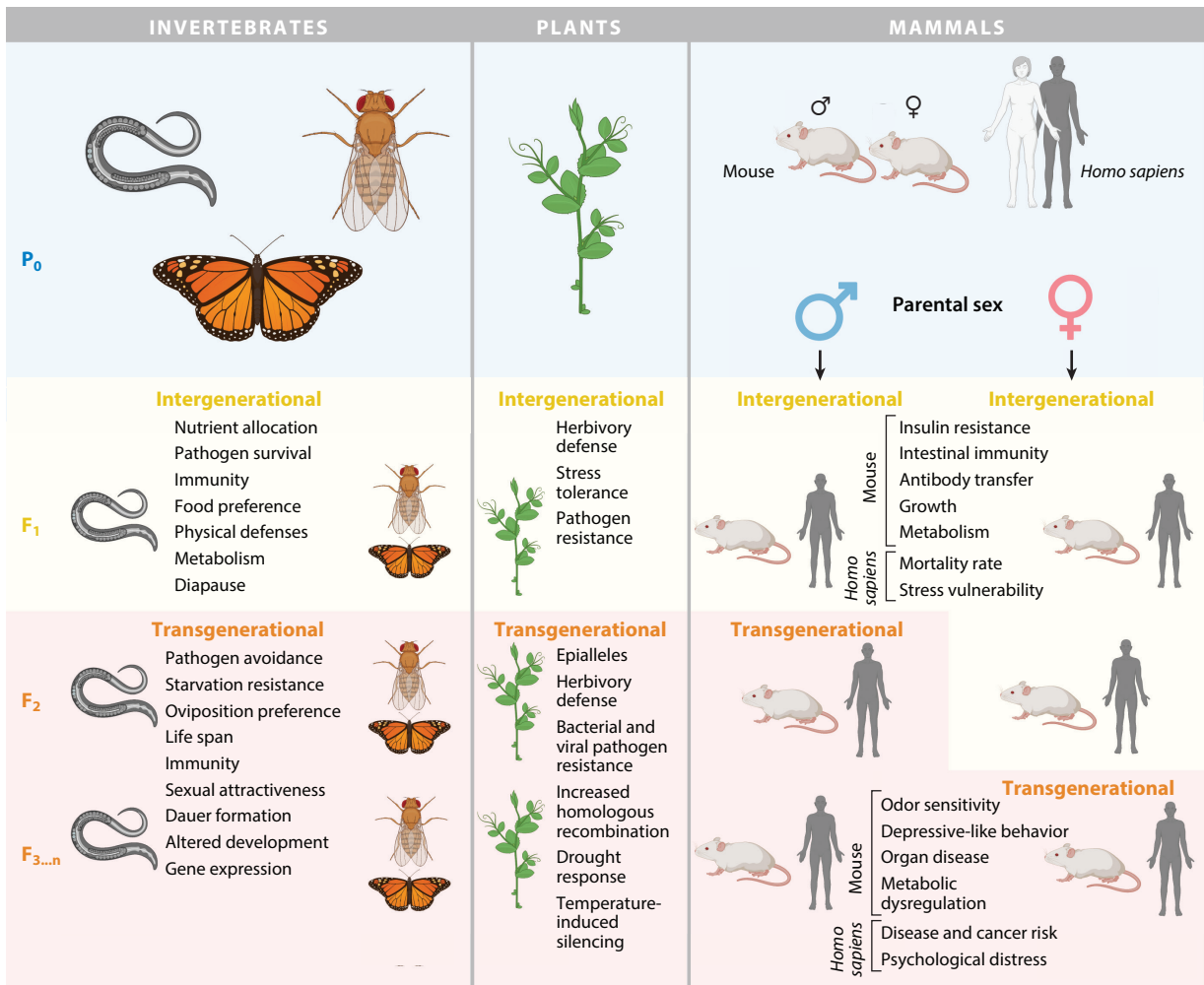


Figure 1

Examples of multigenerational (inter- and transgenerational) inheritance across living systems. In some cases, organisms generate responses that are limited to their own lifetime (intragenerational). In other cases, it may be useful for organisms to generate responses that span multiple generations (intergenerational, lasting one to two generations after exposure, or transgenerational, lasting two or more generations). Inherited changes that persist for only one generation after parental exposure, i.e., in the immediate offspring of the affected parent, fall under the category of intergenerational inheritance. In mammals, a gestating P₀ female carries the F₁ embryo as well as the F₂ germ line (if the F₁ embryo is female); therefore, for the mammalian female line, inherited changes up through the F₂ generation are intergenerational, and only changes in F₃ and beyond are truly transgenerational. Figure adapted from images created with BioRender.com.

Animals have evolved elaborate sensory systems to learn about and respond to their surroundings (Garrity et al. 2010, Sanes & Zipursky 2010). For example, olfactory systems have dedicated mechanisms for responding to environmental odors that reflect either beneficial or dangerous conditions. For instance, following prolonged exposure to specific odorants during starvation, *C. elegans* exhibit a reduced response to these odorants (adaptation) that lasts several hours (Colbert & Bargmann 1995). Epigenetic mechanisms (sRNA pathways and a heterochromatin protein) are employed to sustain the altered response for several hours (Juang et al. 2013). Since

DEFINITION OF TEI AND DISTINCTION FROM INTERGENERATIONAL INHERITANCE

Changes in an organism's environment can lead to inherited changes in its progeny, a phenomenon broadly termed multigenerational inheritance. This can be classified into two categories: intergenerational and transgenerational. The definitions of inter- and transgenerational epigenetic inheritance vary depending on the species and sex, particularly based on the presence or absence of the F1 genetic material at the time of environmental exposure. For all plants and most animal groups, inherited changes that persist for two or more generations after parental exposure to an altered environment are classified as transgenerational inheritance (**Figure 1**).

Transgenerational effects require an epigenetic mechanism of inheritance, usually involving the maintenance of DNA methylation (common in plants) (Zhang et al. 2018) or amplification and reinforcement of other epigenetic modification(s) such as small RNAs and histones (common in invertebrates) (Allshire & Madhani 2018, Berry et al. 2015, Burton et al. 2011, Ciabrelli et al. 2017, Fields & Kennedy 2019, Greer et al. 2010, Heestand et al. 2018, Klosin et al. 2017, Lee et al. 2019, Ouyang & Seydoux 2022, Ouyang et al. 2022, Phillips & Updike 2022, Quarato et al. 2022, Schwartz-Orbach et al. 2020, Seong et al. 2011, Seroussi et al. 2022, Toker et al. 2022). **Intergenerational inheritance, on the other hand, can use nonepigenetic mechanisms such as maternal provisioning of resources (e.g., nutrients, proteins, lipids, and organelles) (Downen & Ahmed 2019, Lockwood et al. 2017, Sato & Sato 2012) as well as long-range signaling from the mother to the developing embryo (Boulinier & Staszewski 2008, Lim et al. 2021). Intergenerational inheritance includes adaptation to or effects of recent environmental changes, whereas transgenerationally inherited effects can serve as a long-term memory of altered environments.**

environments are dynamic, there are also mechanisms in place to reset these changes. In the example of *C. elegans*, animals may need to reset their dampened odorant responses if the odor is now indicative of the presence of food; failure to do so could be detrimental to the worm's survival and fitness. Similarly, positive olfactory associations between neutral odors and food can last for short durations (under 2 h, short-term associative memory) or overnight (long-term associative memory) (Arey & Murphy 2017, Arey et al. 2018, Eliezer et al. 2019, Kauffman et al. 2010, Lakhina et al. 2015) but generally cease quickly, likely reflecting the transient nature of bacterial food sources. Likewise, associative learning in other animals, e.g., olfactory associative learning in flies (Busto et al. 2010) and contextual fear conditioning in mice (Curzon et al. 2009), also lasts for specific, short durations during the exposed animals' lifetime. Therefore, most responses to environmental stimuli and learned associations last for a set period within an organism's lifetime, which is the most appropriate for their environmental circumstances.

Intergenerational Inheritance

There are, however, some behavioral and physiological changes that can be inherited across one or more generations. This type of inheritance is intergenerational (i.e., transmitted only to offspring whose genetic material is present during the parent's lifetime) and is mediated by maternal effects in utero (Gluckman et al. 2008). The F1 progeny may or may not be present at the time of parental exposure. For example, F1 offspring of *C. elegans* exposed to transient early-life starvation exhibit an altered allocation of nutrients despite being conceived after the parental starvation period (Ow et al. 2021). In this case, parent-to-offspring signaling is temporally separated from the parental experience. Intergenerational effects can also be incurred by F1 progeny present in utero as the parent experiences an environmental stimulus. For example, when adult *C. elegans* (containing a developed germ line and, therefore, the genetic material of the F1 generation) are briefly exposed to *Pseudomonas aeruginosa* PA14 (8-h exposure, as opposed to a longer 24-h exposure that induces

transgenerational inheritance; see below) they learn to avoid it and pass on this information to the F1 progeny (Pereira et al. 2020). Similarly, parental microsporidia infection (Willis et al. 2021) and *Pseudomonas vranovensis* exposure (Burton et al. 2020) promote the resistance of the F1 offspring to infection. Parental early-life starvation results in the increased allocation of fats that benefit F1 progeny (Ow et al. 2021). Similar to 8-h PA14 exposure, these effects do not persist in the F2 generation, indicating that these are strictly intergenerational effects. The F1 progeny are present at the time of maternal exposure to the pathogen; therefore, parent-to-offspring signaling can occur concurrently with the changes incurred by the parent due to the exposure.

Changes in odor preference, which involve complex cellular and molecular changes in the nervous system, can also be inherited intergenerationally. For example, in the butterfly *Bicyclus anynana*, exposure to food odors and sex pheromones can cause changes in food or mate preferences, respectively, and these altered preferences persist in the F1 offspring (Dion et al. 2020, Gowri et al. 2019). In the food odor learning study, the butterflies were exposed to the odors as larvae (Gowri et al. 2019), while in the sex pheromone learning experiments they were exposed as adults (carrying the F1 progeny eggs) (Dion et al. 2020). Therefore, the inheritance of altered food and mate preferences might involve distinct inheritance mechanisms.

Extreme morphological changes incurred by the parent in response to the environment can also be intergenerationally inherited. A striking example of intergenerational inheritance of morphological plasticity has been observed in water fleas (*Daphnia*) exposed to predatory fish. In response to predation, *Daphnia* populations develop protective helmets on their heads, and, remarkably, this response is also exhibited by F1 progeny in anticipation of predation (Laforsch & Tollrian 2004).

It is important to determine whether the inherited effects are downstream of the changes in the parental physiology and not direct effects of the stimulus; many studies on intergenerational inheritance have addressed this question and attributed the inherited effects to changes in parental transcription or physiology, such as the maternal provisioning of nutrient proteins (Burton et al. 2020, Hong et al. 2021, Pereira et al. 2020, Perez et al. 2017, Willis et al. 2021). The mechanisms by which physiological changes in the parent result in intergenerationally inherited effects have also been examined in mammals. For example, interleukin 6 (IL-6) produced in response to *Yersinia* infection in female mice causes increased chromatin accessibility in fetal intestinal cells and enhanced intestinal immunity in F1 progeny (Lim et al. 2021).

While the examples discussed above involve inheritance through the maternal line, several cases of intergenerational inheritance occur through the paternal line, usually through epigenetic pathways in sperm. For example, in male mice, specific sRNAs acquired by sperm as they transit through the epididymis are required for the normal implantation and development of F1 embryos (Conine et al. 2018). In *C. elegans* sperm, sRNA signals might underlie a memory of male-specific gene expression (Conine et al. 2013). The mechanisms underlying inherited effects are easier to dissect in these examples as they are not confounded by the influence of factors in the maternal uterine environment.

Transgenerational Inheritance

According to Weismann's theory, somatic changes are not expected to be inherited through the germ line; therefore, transgenerational inheritance is counterintuitive and intriguing. Inherited effects that persist beyond the F1 generation (F2 generation for inheritance through the female line in mammals) represent spectacular deviations from Weismann's theory, and they are termed transgenerational effects. Most reported transgenerational phenomena involve heritable epigenetic mechanisms and are referred to as TEI.

Over the past few decades, multiple examples of TEI have been reported, particularly in invertebrate model systems. Odors are potent cues for transgenerational inheritance as they are often

informative about environmental conditions, including the presence of food, mates, pathogens, and predators. For example, *Drosophila* exposed to predatory wasps lay their eggs on ethanol to protect their progeny from wasp infection (Bozler et al. 2019). This adaptive preference for oviposition on ethanol is inherited for five generations (Bozler et al. 2019). In *C. elegans*, exposure to attractive odorants in early life produces imprints that regulate behavior upon exposure to the corresponding odorants (Remy 2010); there may be a transgenerational effect if exposure to the odor lasts for at least four generations, although the molecular mechanism underlying such an effect has not been identified (Remy 2010). Odor-related stimuli may produce transgenerational effects in mammals as well: Odor fear conditioning in male mice by pairing the smell of acetophenone to aversive cues can result in increased sensitivity to acetophenone in the F1 and F2 generations, through DNA methylation changes in a specific olfactory receptor (Dias & Ressler 2014).

Starvation is a powerful stressor that may induce both short-term and long-term adaptive responses, including transgenerational responses. *C. elegans* likely encounters fluctuating nutrient levels (Schulenburg & Félix 2017) and bouts of starvation in their natural environment. Therefore, inheriting information about the availability of nutrients is likely beneficial in some circumstances, particularly with severe or long-lasting nutrient deprivation. For example, extended early-life starvation of *C. elegans* results in increased starvation resistance up to the F2 generation (Jobson et al. 2015).

Pathogen exposure is a constant threat that may also induce variable levels of adaptive responses, depending on the duration of exposure and the degree of pathogenicity. *C. elegans* is a free-living nematode that encounters a wide range of microbial species in its natural habitats (Frézal & Félix 2015, Schulenburg & Félix 2017). Certain bacterial genera, e.g., *Pseudomonas*, can serve as a source of nutrition but can also become pathogenic under specific environmental conditions, such as at higher temperatures (Moore et al. 2019). It is likely advantageous for *C. elegans* to be able to respond to the presence of (temporarily) lethal bacteria such as pathogenic *Pseudomonas* and pass on beneficial behavioral strategies to their progeny for the duration of the pathogenic conditions. *C. elegans* exposed to the pathogenic bacterium *P. aeruginosa* (PA14) for only 4 h can learn to avoid it, but this response only lasts through the exposed generation (Zhang et al. 2005), while 24 h of exposure induces a learned avoidance that is passed on to its progeny for four successive generations—true transgenerational inheritance of learned avoidance (Kaletsky et al. 2020; Moore et al. 2019, 2021a, 2021b). This adaptive transgenerational response is induced by a specific sRNA made in the PA14 bacteria called P11 (Kaletsky et al. 2020). P11 sRNA is processed by sRNA pathways in the *C. elegans* mother, and signals from the germ line regulate the inheritance of the response (Kaletsky et al. 2020). Altered gene expression in neurons enables the learned behavior in the mother and the progeny, and the neuronal changes driving the avoidance behavior in the progeny are distinct from those underlying avoidance in the mother (Kaletsky et al. 2020; Moore et al. 2019, 2021a).

The difference between an acute response in the P0 generation after only 4 h of exposure (Zhang et al. 2005), an intergenerational response after 8 h (Pereira et al. 2020), and a transgenerational response upon 24 h of exposure to this lethal pathogen (Moore et al. 2019) suggests that different lengths of exposure induce separate molecular machineries that set the duration of response. Moreover, a less lethal strain of *P. aeruginosa* called PA01 requires multiple generations of exposure to induce any response (Palominos et al. 2017), suggesting that the severity of pathogenicity helps set the generational duration of response. This set of studies provides a striking example of the tuning of generational responses by environmental conditions.

Intragenerational adaptation, intergenerational inheritance, and transgenerational inheritance employ distinct cellular and molecular mechanisms. Intragenerational adaptive changes such as sensory adaptation, e.g., odor adaptation, can be actuated by plastic changes at the level of

synapses and circuits (Wilson 2013). Intergenerational inheritance involves long-range signaling from the mother to the embryo through hormones, cytokines, nutrients, and other signals. For example, *C. elegans* subjected to osmotic stress signal through insulin-like peptides to the maternal germ line to regulate glycerol metabolism in F1 progeny, thereby conferring resistance to osmotic stress (Burton et al. 2017). In the *Bombyx* silkworm, a hormone from the suboesophageal ganglion of the mother regulates diapause in F1 progeny (Hasegawa 1957). Transgenerational inheritance often requires epigenetic mechanisms like amplificatory sRNA pathways, the reinforcement of histone modifications, and the maintenance or incomplete erasure of DNA methylation marks (Fitz-James & Cavalli 2022, Liberman et al. 2019).

MULTIGENERATIONAL INHERITANCE ACROSS LIVING SYSTEMS: COSTS AND BENEFITS

Multigenerational inheritance can be adaptive or costly, depending on an organism's life history and environment. Inter- or transgenerational inheritance might serve as (a) mechanisms to inform progeny about their environment and confer adaptive mechanisms used by the parent to the progeny (Kaletsky et al. 2020, Moore et al. 2019, Ow et al. 2021) and (b) mechanisms to silence potentially deleterious genomic elements (such as retroviruses) across generations (Brennecke et al. 2008). On the other hand, if an organism's environment is different from ancestral environments, then inherited information can be costly to maintain and may be mismatched with respect to the environment (Figure 2) and, therefore, be maladaptive.

Plants harbor natural epialleles—genetic loci with identical DNA sequences but varying epigenetic marks—that have persisted across hundreds of generations (Weigel & Colot 2012). Epialleles

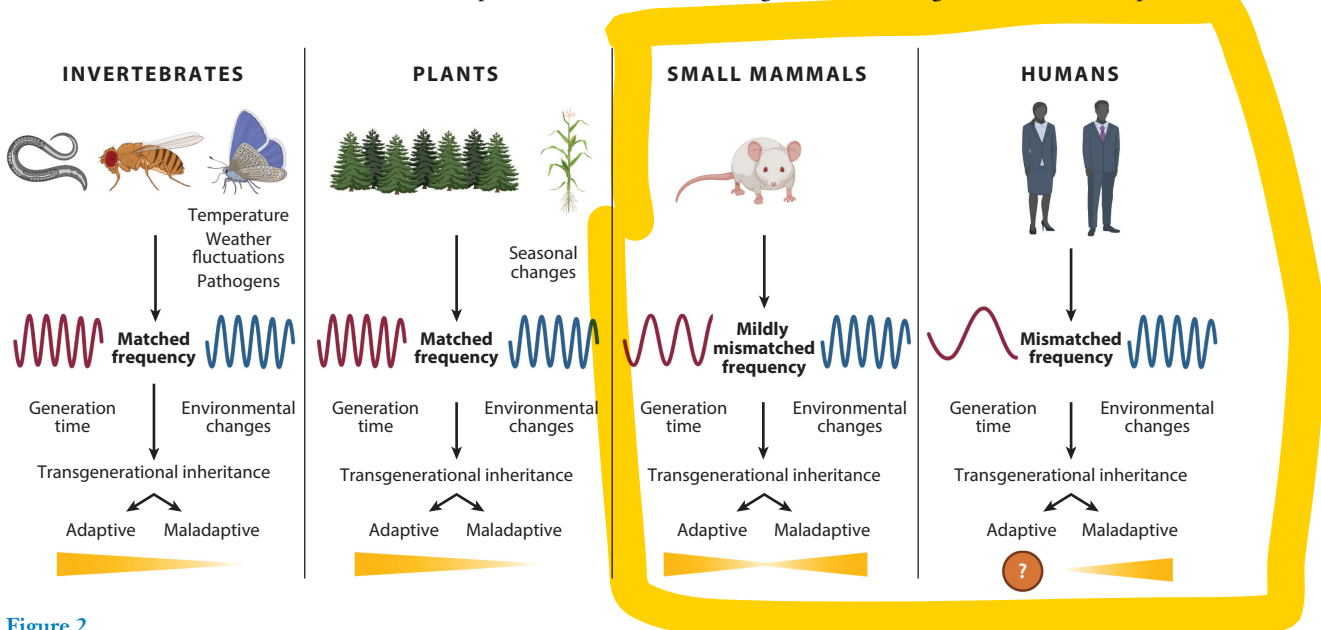


Figure 2

Multigenerational inheritance can be adaptive or costly depending on the environment and organism life histories. If environmental changes are less frequent compared to generation times (e.g., for fast-reproducing organisms like plants and invertebrates), then inherited changes might have adaptive value. If the environment changes at a higher frequency than the generation time (e.g., mammals), inherited effects might be mismatched with respect to the environment and, therefore, are more likely to be maladaptive. Figure adapted from images created with BioRender.com.

are often located near methylated transposons and may have arisen by the spreading of methyl marks from the silenced transposable element into a nearby gene promoter. Since transposon silencing is critical for normal plant development, the methylated and, therefore, silenced state of the transposon is permanently maintained across both the transposable element and the adjacent gene comprising the epiallele (Baduel & Colot 2021).

However, not all inherited epigenetic loci are permanently maintained, as in epialleles; there are examples of stimulus-induced transgenerational inheritance in the plant kingdom that lasts only for a set number of generations. For example, in *Arabidopsis*, chemicals such as jasmonic acid provide protection against caterpillar herbivory in the exposed generation and two additional generations (Rasmann et al. 2012). Similarly, exposure to the pathogen *Pseudomonas syringae* can confer transgenerational acquired resistance to *Arabidopsis* populations, and these signatures persist at the epigenomic level three generations after exposure (Stassen et al. 2018). These environmental stressors are usually temporary in nature, so resetting the inherited effects within a set number of generations is likely beneficial.

The transgenerational inheritance of environment-induced epigenetic marks is an evolutionarily favorable strategy in plants for the following reasons: (a) Plants are sessile organisms that must adapt to changing environments (Miryeganeh & Saze 2020). Epigenetic changes that are heritable provide a flexible form of adaptation that can be reversible, unlike hardwired changes in DNA sequence; these reversible epigenetic changes allow the plant and its progeny to respond to environmental stresses that could persist for multiple generations. Epigenetic plasticity through generational time also allows the plant's progeny to reset to the previous epigenetic state when the environmental stress is alleviated. (b) The plant germ line differentiates relatively late in the life cycle (Bouyer et al. 2017), when the somatic cells exposed to environmental factors have already incurred stimulus-induced epigenetic changes; as a result, these changes can be inherited by the developing gametes as they form. (c) TEI has mostly been observed and studied in model annual plants (Lang-Mladek et al. 2010, Rasmann et al. 2012), but there is potential for TEI to also occur in plants with alternate life history traits, such as perennial plants. Because perennials are longer lived, they can potentially integrate information about the environment over many years and therefore harbor potentially heritable epigenetic changes that are representative of long-term patterns in the environment. It will be interesting to examine if perennial plants employ shared or distinct strategies and mechanisms of TEI compared to annual plant species.

Many examples of transgenerational inheritance in invertebrates are also induced by environmental stimuli, and these are reset within a few generations. For example, *Pseudomonas*-induced learned avoidance in *C. elegans* lasts for four generations after exposure and resets in the fifth (Kaletsky et al. 2020; Moore et al. 2019, 2021a). Most *Pseudomonas* species are detrimental to worms (Samuel et al. 2016), and worms can encounter *Pseudomonas* in their environment for multiple generations (approximately days to weeks for worms). Therefore, it may be worth generating a TEI signal of avoidance in response to *Pseudomonas* exposure. However, since many *Pseudomonas* species are likely to also serve as a source of nutrition under other environmental conditions (Moore et al. 2019, Samuel et al. 2016), erasing the avoidance signal once the immediate pathogenic threat has passed is likely beneficial. Mutations in histone modification pathways can also result in transgenerational physiological effects, suggesting that histone modifications might normally endogenously regulate physiological processes and their inheritance. One of the most striking examples of this is the transgenerational inheritance of extended life span by wild-type progeny of mutants of the COMPASS histone regulatory complex (Greer et al. 2010, Lee et al. 2019).

Mammals, on the other hand, often have longer generation times (e.g., a few months for mice to several decades in humans). Retaining information about an ancestral environment may not

make sense in such cases since stimuli such as starvation or pathogen exposure are unlikely to persist for multiple generations. The environmental challenges that mammals face are likely to be resolved long before transgenerational responses could provide any potential benefit to the progeny. Intergenerational inheritance, which lasts for one to two generations (one for the male line and two for the female line in mammals), is therefore more likely to be a beneficial adaptive strategy in mammals. Consistent with this notion, there are multiple examples of adaptive intergenerational effects in mammals, e.g., the inheritance of antibodies from parent to offspring (Boulinier & Staszewski 2008) and oxytocin-mediated intergenerational transmission of maternal behavior in rodents (Carcea et al. 2021), and maternal glucocorticoids resulting in adaptive growth in F1 progeny in red squirrels (Dantzer et al. 2013). In each of these cases, the second generation benefits immediately from changes conferred to it by the mother. For hibernating mammals such as squirrels, it would be interesting to investigate if hibernation, which results in modulation of the epigenome (Alvarado et al. 2015), can reset epigenetic changes incurred by the parent and prevent inheritance.

CONVERSION OF ENVIRONMENTAL INFORMATION TO INHERITANCE SIGNALS

Inheritance of environmental memory usually requires a systemic response whereby the tissue responsive to the environment communicates with the germ line to orchestrate inheritance. An understanding of multigenerational inheritance requires characterization of this organism-wide response.

The intestine is subjected to various environmental assaults and can sense changes in nutrients and pathogens (Balla & Troemel 2013, Qi et al. 2017). The *P. aeruginosa* strain PA14 infects the intestine of *C. elegans*, where PA14's sRNA (P11), which drives the inheritance of learned avoidance to PA14, is transported from the intestine to the germ line, where inheritance signals are generated (Kaletsky et al. 2020, Moore et al. 2019). Another instance of intestine-to-germ line signaling has been observed in *C. elegans* mutants with intestinal bloating (Hong et al. 2021). These mutants exhibit altered germ line H4 Lys8 acetylation, but the exact nature of the intestine-germ line signal is yet to be identified.

Neurons provide another window into diverse environmental signals, including odors, visual signals, and other sensory cues. Epigenetic signals generated in neurons can be communicated to the germ line, which has been revealed through studies of synthetic systems and mutant analysis. For example, ectopic expression of double-stranded RNA to GFP in *C. elegans* neurons can cause GFP silencing in multiple tissues, including the germ line (Devanapally et al. 2015). Similarly, in mutant worms lacking the double-stranded RNA-binding protein RDE-4, overexpression of RDE-4 in neurons can alter germ line sRNAs and chemotaxis behavior (Posner et al. 2019). However, whether such neuron-to-germ line signaling occurs in wild-type animals in response to natural environmental effects on neuronal sRNAs (as opposed to experimental manipulation of sRNAs in a specific tissue in whole-body mutant animals) is not yet known.

In addition to *C. elegans*, the molecular mechanisms and genes affected in other species by some environmental stimuli are known. For example, transgenerational inheritance of odor sensitivity upon odor fear conditioning in male mice involves hypomethylation of an odorant receptor *Olfir151* in sperm of the mice exposed to the conditioning (Dias & Ressler 2014), but how activation of this receptor in neurons results in a hypomethylated state at the corresponding gene locus in germ cells (specifically sperm) is not yet understood. However, in some cases, how neurons communicate with the germ line is known. Neuropeptide signaling is a mode of long-range communication across tissues (Beets et al. 2022, Frakes et al. 2020, Ramachandran et al.

2021, Ripoll-Sánchez et al. 2022, van den Pol 2012), and modulation of neuropeptide-mediated neuron-to-germ line signaling can serve as a mechanism of the transgenerational inheritance of neuronal signals. For example, exposure to predatory wasps causes Neuropeptide F inhibition in the fly brain, which, in turn, triggers germ line caspases that are required for the transgenerational inheritance of altered food preference (Bozler et al. 2019). In *C. elegans*, increased population density information is perceived by neurons through pheromones and transmitted to the germ line through TGF- β signaling (Perez et al. 2021). It will be interesting and important to characterize the repertoire of neuron-to-germ line signaling mechanisms employed in diverse contexts, particularly in response to environmental stimuli.

TYPES OF ENVIRONMENTAL MEMORY: STIMULUS-TO-EFFECT RELATIONSHIPS

The nature of multigenerational inheritance of a phenotype depends on the environmental stimulus that triggers inheritance. The number of generations of exposure to a stimulus required to generate a heritable response, and the number of generations for which the response is inherited, depends on the stimulus and its effects on the exposed parent. For example, the *P. aeruginosa* strains PA14 and PA01 differ in their pathogenicity to *C. elegans* (Tan et al. 1999). Worms have distinct strategies to respond to these pathogens, and exposure to PA14 and PA01 produces distinct acute and transgenerational effects. PA14 is a potent pathogen, killing worms within a few days of exposure (Tan et al. 1999); therefore, exposure to this deadly pathogen constitutes environmental information worth remembering and subsequently passing on to progeny upon a single generation of exposure. With a single exposure to PA14 for only 24 h, worms can learn to avoid the pathogen, and this avoidance memory is passed on to four generations of progeny (Kaletsky et al. 2020, Moore et al. 2019). The learned avoidance provides a valuable strategy for worms, as this allows them to forage for other noninfectious bacterial food sources required to provide the nutrition needed to promote both survival and fitness. By contrast, persistent and multigenerational exposure is required to generate a response to the far less pathogenic PA01, and the transgenerational response itself is extremely mild: A very small fraction of worms enter a nonreproductive stress-resistant state called dauer (Palominos et al. 2017). Specifically, when worms are constantly exposed to PA01 over at least two successive generations, about 4–5% of the F3 progeny of the last exposed generation enter the dauer state upon pathogen reexposure (Legüe et al. 2022, Palominos et al. 2017). Since PA01 is a milder pathogen, it is possible that worms delay the production of a transgenerational signal until they have been exposed for multiple generations, which could then indicate the lack of other nutritious food options and trigger entry into diapause. Inducing the nonreproductive dauer state only in a small percentage of worms might be adaptive, as it protects a part of the population from infection while maintaining reproductive ability in 95% of the population. The difference in transgenerational responses may reflect the worm's ability to tune its responses to varying environmental threats.

The duration of exposure to a stimulus might also determine whether there are heritable effects. For example, upon a 4-h exposure to PA14, *C. elegans* hermaphrodites both learn to avoid PA14 and increase the rate of mating with males, two acute responses to pathogen exposure (Wu et al. 2023). [Mating is a strategy to increase genetic diversity, and it is induced under a variety of stressful conditions, including starvation, heat, and aging (Plesnar-Bielak et al. 2017).] While the memory of avoidance induced by a 4-h exposure is not inherited, a 24-h exposure to PA14 induces learned avoidance that is inherited for four generations (Kaletsky et al. 2020, Moore et al. 2019). These observations suggest that a threshold duration might be required for the generation of an inherited signal. The inherited avoidance of PA14 requires the PA14 sRNA P11 to be absorbed

in the intestine and transported and processed in the germ line (Kaletsky et al. 2020), which may take longer than the avoidance response triggered by a 4-h exposure and could explain why the longer exposure is necessary to induce a transgenerational response.

Multigenerational inheritance might be triggered only upon exposure to a stimulus in critical periods of an organism's life cycle. For example, *C. elegans*' learned avoidance to PA14 upon 24-h exposure requires a mature germ line (Moore et al. 2019) and therefore cannot take place earlier than the late larval or adult stages. By contrast, the effects of starvation in early larval stages in *C. elegans* manifest as physiological changes later in life, some of which are transgenerationally inherited (Jobson et al. 2015, Vogt & Hobert 2023). The starvation response involves changes in endogenous sRNAs (Rechavi et al. 2014), which might have higher perdurance (and can therefore underlie phenotypes in later developmental stages) relative to exogenous sRNAs like P11 that drive the inherited response to PA14 (Kaletsky et al. 2020). Physiological states corresponding to specific developmental stages might also facilitate the production of heritable signals.

Similar phenotypes induced by distinct environmental stimuli might differ in heritability. For example, the increased mating frequency observed in *C. elegans* in the parental generation upon acute (4-h) exposure to *Pseudomonas* (Wu et al. 2023) is similar to the multigenerational heat stress induction of mating that is transgenerationally inherited (Toker et al. 2022). An increase in mating can diversify the gene pool to confer enhanced adaptability in the face of environmental stress. Distinct environmental stressors might select for an intragenerational response or a transgenerational response, depending on the evolutionary context.

TEI IN MAMMALS: LOGIC AND CRITICAL EVALUATION

Does TEI Make Sense in Mammals?

The conservation of transgenerational epigenetic phenomena and mechanisms across plants and invertebrates speaks to the importance of these processes in evolution. This is supported by the variety of TEI examples discovered to date that have a demonstrable benefit for the organism and its progeny, including pathogen and environmental stress resistance, behavioral adaptations, and metabolic/life span effects (Bozler et al. 2019, Jobson et al. 2015, Klosin et al. 2017, Moore et al. 2019). These diverse organisms have rapid generation times relative to mammals, including humans, such that TEI as an adaptive response makes sense. For example, a heat wave, drought, or the presence of a pathogen may persist in the environment on a temporal scale that lasts multiple generations for a fly, worm, or plant. Preparing future generations of these organisms for these potential challenges may provide key survival advantages.

But does TEI make sense in mammals? Most mammals, from mice to humans, have longer generation times than most invertebrate organisms. As a result, the environmental challenges a mammal faces may be resolved long before transgenerational responses could provide any potential benefit to the progeny (Figure 2). Experimental limitations aside, perhaps this explains why so many of the multigenerational phenomena identified in mammals [and in some fish (Kelley et al. 2021)] are in fact not transgenerational but rather intergenerational, where it could benefit the parent to alter the epigenetic landscape of the progeny in utero but not to generations beyond that. For example, many phenomena are due to events that occur in F1 during gestation—e.g., enhanced fetal intestinal immunity by exposure to IL-6 (Lim et al. 2021), oxytocin effects on offspring (Carcea et al. 2021), or antibody transfer from the mother to the fetus (Boulinier & Staszewski 2008)—and thus are intergenerational effects. But true TEI phenomena are rare in mammals—and this may have been evolutionarily selected (Kelley et al. 2021). That is, molecular mechanisms of TEI might primarily function to silence potential generational memories. Why



should grandprogeny remember conditions that grandparents experienced many years ago but may no longer exist?

The molecular components that regulate TEI, including sRNA/RNAi regulators, chromatin modifications, and methylated DNA, are highly conserved across diverse organisms (Fitz-James & Cavalli 2022), suggesting that TEI is mechanistically possible in humans. However, most cases of TEI in mammals might be a maladaptive by-product of possessing conserved molecular machinery. For example, differential DNA methylation regions in sperm have been associated with reproductive and kidney diseases in male F3 progeny of rats exposed to toxic chemicals such as vinclozolin (Anway et al. 2006). Exposure to pesticide mixtures and plastic-derived chemicals during gonadal sex determination in mice can result in F1–F3 progeny with specific differentially methylated regions in sperm and higher incidence of a variety of diseases associated with the reproductive system (Manikkam et al. 2012, 2013). The inheritance of these disease states might be a maladaptive by-product of possessing DNA methylation machinery. Similarly, the memory of an odor-conditioned response to stress in grandprogeny (F2) of the conditioned generation (Dias & Ressler 2014) is somewhat alarming, as it suggests that animals might respond to cues that were not present in their own lifetimes, and therefore could be deleterious. In another example of maladaptive inheritance in mice, a chronic social instability paradigm increases anxiety and defects in social interactions that are transmitted to F2 female offspring of the F1 male progeny of stressed fathers (Saavedra-Rodríguez & Feig 2013).

Several instances of maladaptive TEI have also been reported in humans. In individuals conceived during a brutal winter famine, the Dutch Hunger Winter, the adverse environment in utero due to the mother's starvation resulted in increased disease risk later in life (Heijmans et al. 2008), particularly metabolic diseases that may result from a shifting of the metabolic set point in utero, as well as psychiatric disorders; fascinatingly, later generations exhibited increased disease risk as well, suggesting a maladaptive transgenerational effect (Bleker et al. 2021). Similarly, populations in Överkalix, Sweden, experienced chronic oscillations in under- and overnutrition in the nineteenth century, and longitudinal studies have been carried out on these populations and subsequent generations (referred to as the Överkalix cohort) to examine transgenerational effects. For example, the paternal grandfather's access to food determined the mortality of male grandchildren, and prepuberty overnourishment in paternal grandfathers caused elevated mortality, including elevated cancer risk in male grandprogeny (Vågerö et al. 2018). Additional studies from population databases in Sweden revealed that individuals who experienced parental death during childhood have offspring with lower birth weight (Hiyoshi et al. 2021). However, it is important to note that the effects of generational trauma might be confounded by stress, lack of parental care, and other sociocultural experiences.

Maladaptive TEI processes in mammals might be parallel to TEI mechanisms in invertebrates that result from genetic perturbations. Inheritance mechanisms in these mutants might be a so-called epigenetic hangover from having a mutant background that alters epigenetic pathways that are normally in place to erase potentially long-lasting signals. For example, genetically wild-type progeny of outcrossed *C. elegans* COMPASS complex mutants exhibit longer life spans (Greer et al. 2010), and offspring of *rde-4* mutants exhibit altered chemotaxis (Posner et al. 2019). Wild-type animals do not show these life span and chemotaxis phenotypes, suggesting that there are mechanisms normally in place to prevent the possibility of inheriting these types of signals. This contrasts with the mechanisms of TEI in wild-type animals in response to pathogens or starvation (Hibshman et al. 2016; Jobson et al. 2015; Kaletsky et al. 2020; Moore et al. 2019, 2021a) that might help animals survive for multiple generations (although they might share some common molecular mechanisms). Since it is potentially maladaptive to retain transgenerational memories, overwriting environmental information is adaptive for mammals, including humans, and dedicated



molecular mechanisms may have evolved for resetting these memories. It will be important to identify these mechanisms in mammals in order to develop treatments to reset inappropriate or maladaptive transgenerational responses.

Critical Evaluation of Transgenerational Effects



How widespread is transgenerational inheritance? Evidence for transgenerational inheritance, particularly in mammals, must be critically evaluated. To begin, only effects observed F2 onwards (paternal line) or F3 onwards (maternal line) should be categorized as TEI; others should be correctly categorized as intergenerational inheritance. As discussed above, these two forms of inheritance (inter- and transgenerational inheritance) might be very different mechanistically and in terms of their ecological significance. Therefore, correct nomenclature is critical for accurately contextualizing new discoveries in this rapidly evolving field. This nomenclature issue extends to the lack of consistent phenotypes in multiple generations; if inherited effects are observed in one of the sexes in specific progeny generations, and if opposite effects are observed between generations, further investigation should be done to address the biological significance of those observations. For example, if a phenotype is only reported in males of the one generation and females of another generation, or a new phenotype is seen in a much later generation (Boscardin et al. 2022, Franklin et al. 2010), and all show low statistical significance, then whether this is truly a transgenerational effect should be evaluated. It is important to determine not only what is a TEI phenotype but also whether these effects are reproducible and significant in order to understand what we might do to interrupt maladaptive TEI effects.

One of the points of using model systems—as opposed to relying solely on the interpretation of human historical natural experiments—is to be able to carry out experiments under conditions that will make the interpretation of the results clear, allowing the work to be grounded in informed hypotheses and predictions. However, some mouse studies of TEI suffer from small sample sizes and low significance (Mitchell 2013). It is particularly important that we have a firm understanding of TEI and its possible limitations in mice, the leading mammalian model system (but still a model organism), before we can extrapolate such findings to human biology. Nevertheless, the possibility of translating some of the in-depth molecular findings from invertebrate studies to mammals, and using this information to understand how we might reduce the deleterious effects of transgenerational and historical trauma, is exciting.

CONCLUSIONS

Organismal responses to environmental stimuli span widely different timescales; while some responses last for minutes or hours in an organism's lifetime, others may persist for generations (Baugh & Day 2020). The duration and severity of environmental inputs may determine the temporal dynamics of a response; that is, the type of input may determine whether the organism's response is intra-, inter-, or transgenerational. The generational scale of a response might be adaptive or deleterious, depending on an organism's life history traits.

To generate heritable signals, tissues exposed to environmental stimuli must communicate with the germ line, which orchestrates inheritance. Identifying the tissue that transduces environmental stimuli to generate signals that establish communication with the germ line is critical in achieving a complete understanding of multigenerational inheritance. Some experimental reports on multigenerational inheritance may not reflect the inherited responses to an environmental stimulus, but instead they may describe runaway inheritance phenomena in mutants and transgenics where mechanisms that normally erase memories and restrict their inheritance have gone awry.

For organisms with relatively longer generation times such as mammals, intergenerational inheritance might have more adaptive value than transgenerational inheritance; observed transgenerational inheritance phenomena might instead be the result of maladaptive mechanisms. While many examples of robust transgenerational inheritance have been characterized in invertebrates (Bozler et al. 2019, Gammon et al. 2017, Greer et al. 2010, Hibshman et al. 2016, Jobson et al. 2015, Kaletsky et al. 2020, Klosin et al. 2017, Mondotte et al. 2020, Moore et al. 2019, Perez & Lehner 2019, Rechavi et al. 2014, Wang et al. 2022), critical examination of experimental results and concepts in mammalian TEI paradigms is required before they can be extrapolated to humans.

The field of TEI is exciting and fast-moving, with new developments in multiple model systems appearing with increasing frequency. In addition to uncovering the molecular mechanisms that mediate TEI phenomena, considering the contexts in which environmental signals result in meaningful adaptive responses is useful to understand how TEI might be initiated, how a TEI response might persist, and how we might block maladaptive responses to benefit humans.

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